

ORF307

Piecewise Linear Optimization

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Today's topics

- Vector norms
- Piecewise linear optimization
- Turning vector norm problems to LPs
- Sparse signal recovery
- Support vector machines

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- **Vector norms**

- **Euclidean norm (or L_2 norm)**

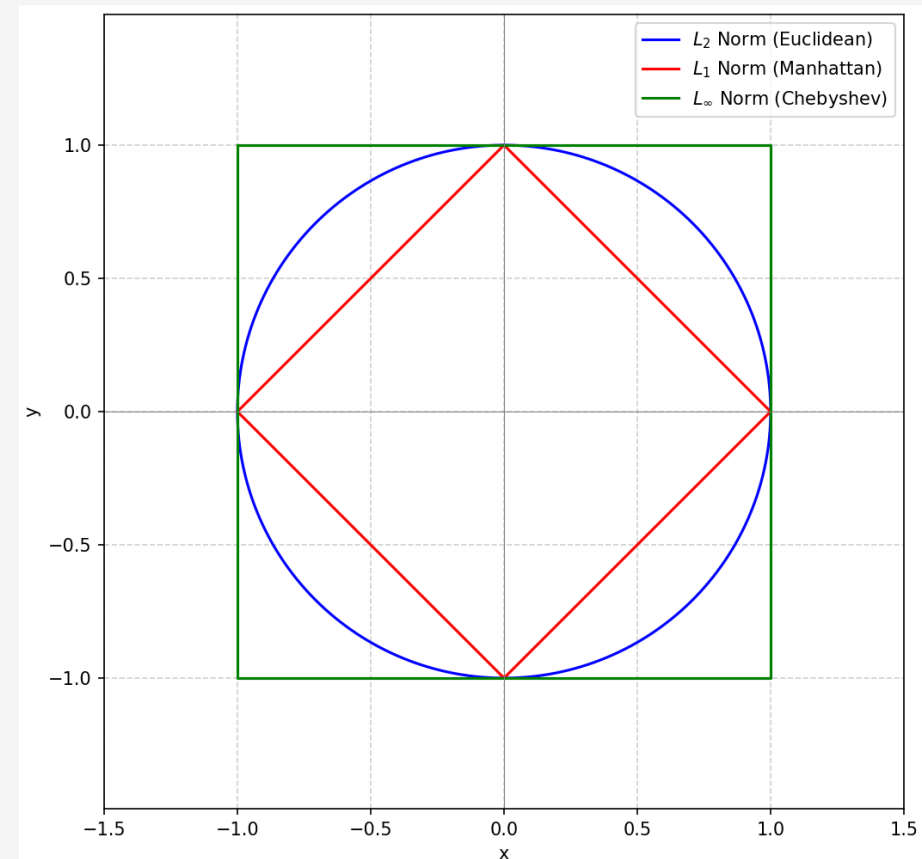
$$||x||_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

- **1-norm (or L_1 or Manhattan norm)**

$$||x||_1 = \sum_{i=1}^n |x_i|$$

- **∞ -norm (or L_∞ norm)**

$$||x||_\infty = \max_{i=1\dots n} \{|x_i|\}$$



Data fitting example

Dataset: $D = \{(z_i, f_i) \in \mathbb{R}^2 : i = 1, 2, \dots, m\}$

Fit a linear function: $\hat{f}(x) = x_1 + z x_2 = (1, z) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

Find the best (x_1, x_2) such that

$$f_i \approx \hat{f}(x_i) = x_1 + z_i x_2, \quad \forall i = 1, 2, \dots, m$$

In matrix-vector form:

$$Ax = b \Leftrightarrow \begin{bmatrix} 1 & z_1 \\ 1 & z_2 \\ \vdots & \vdots \\ 1 & z_m \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_m \end{pmatrix}$$

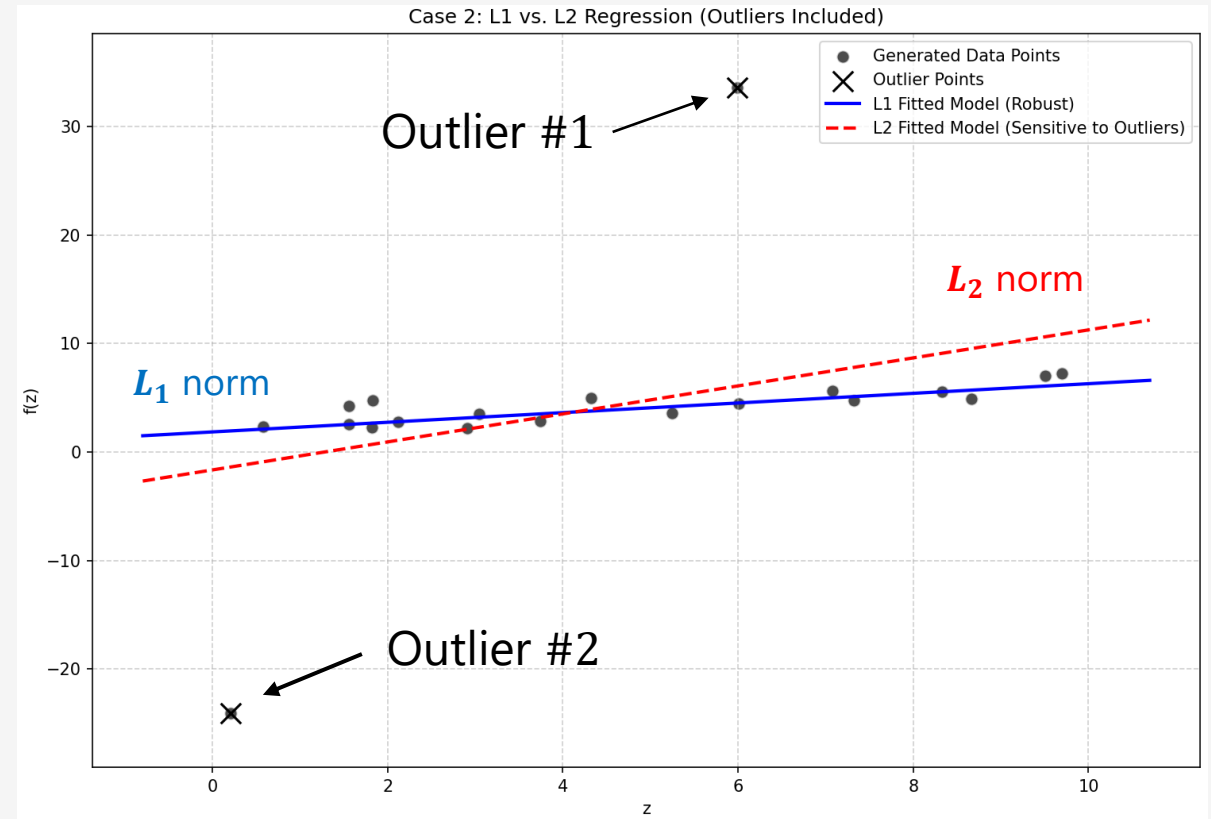
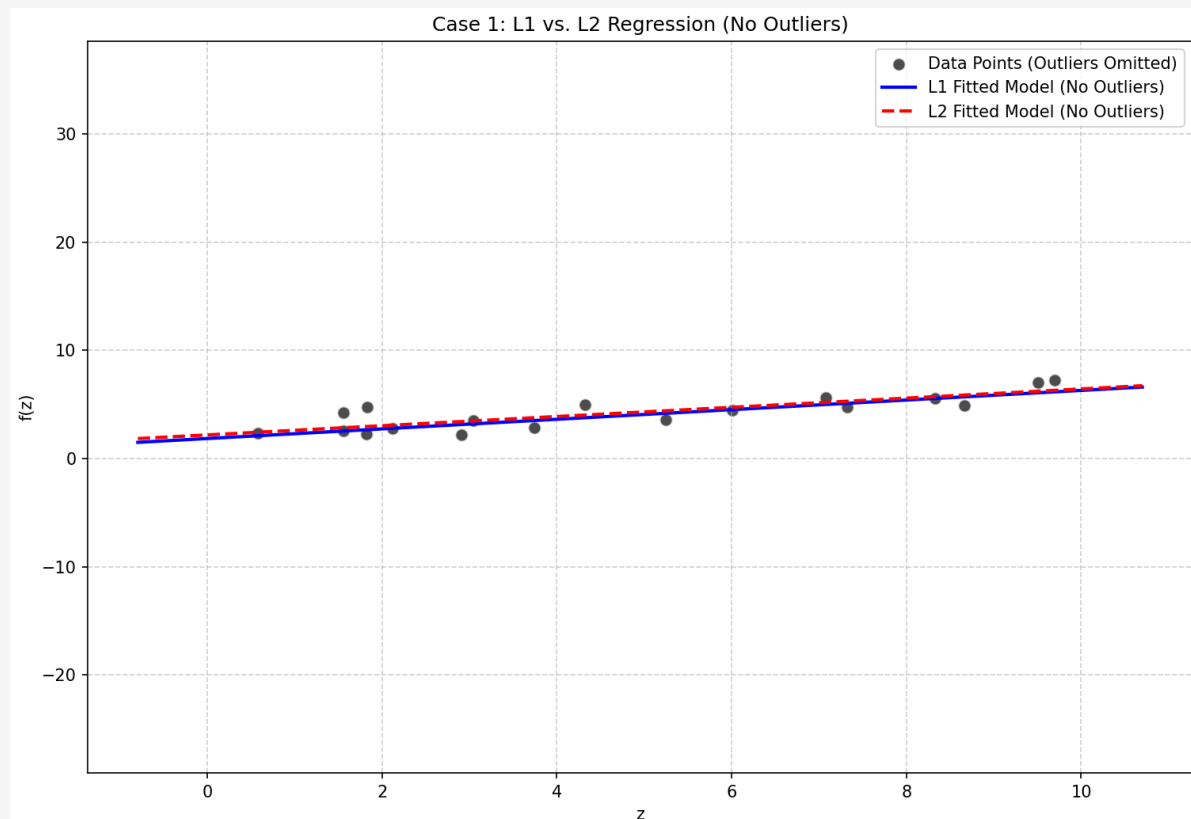
Using L_1 norm (**L**east **A**bsolute **D**eviation):

$$\min \sum_{i=1}^m |Ax - b|_i = \|Ax - b\|_1$$

Advantage: x^* is more robust to outliers

Disadvantage: no closed-form solution

The Optimization problem can be converted to an LP



The L_1 model is willing to tolerate some larger residuals (caused by the outliers) as long as it can make most of the smaller residuals as close to zero as possible. That's why the L_1 model is less sensitive (more robust) to the presence of outliers in the data.

From many solutions to a unique solution

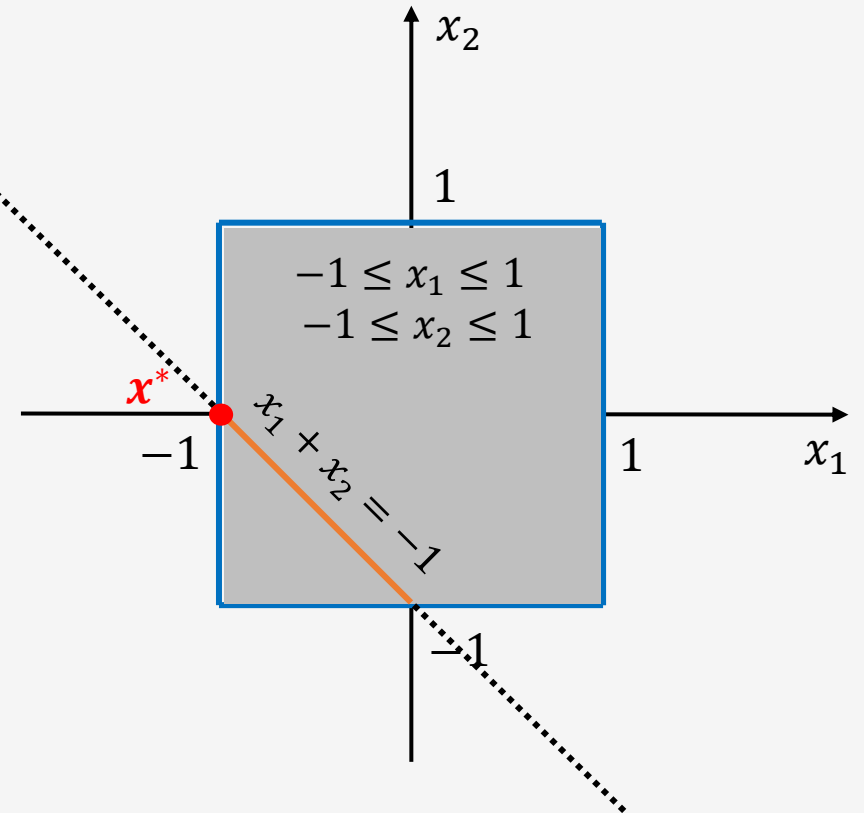
$$\begin{array}{ll}\min & x_1 + x_2 \\ \text{s.t.} & -1 \leq x_1 \leq 1 \\ & -1 \leq x_2 \leq 1 \\ & x_1 + x_2 = -1\end{array}$$

LP with many solutions.

(all points among points
 $(-1,0)$ and $(0,-1)$ are
optimal solutions)

$$\begin{array}{ll}\min & x_1 + x_2 \\ \text{s.t.} & \|x\|_\infty \leq 1 \\ & x_1 + x_2 = -1\end{array}$$

**It is not an LP, but it can
be converted to an
equivalent LP that has a
unique solution at x^***



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- **Piecewise linear optimization**

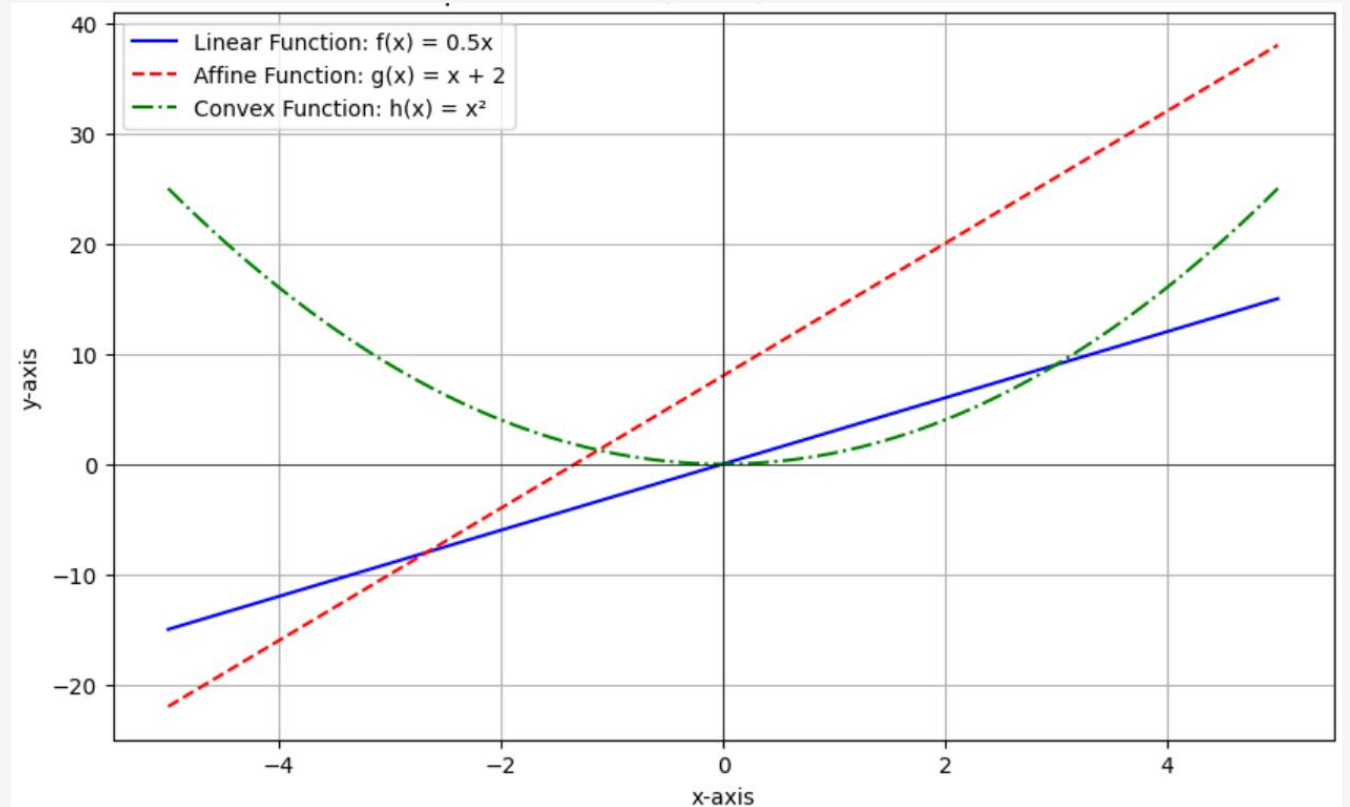
Linear, affine and convex functions

Linear function: $f(x) = a^T x$

Affine function: $g(x) = a^T x + b$

Convex function: $\forall x, y \in R^n, \forall \lambda \in [0,1],$

$$h(\lambda x + (1 - \lambda)y) \leq \lambda h(x) + (1 - \lambda)h(y)$$



Convex piecewise linear functions

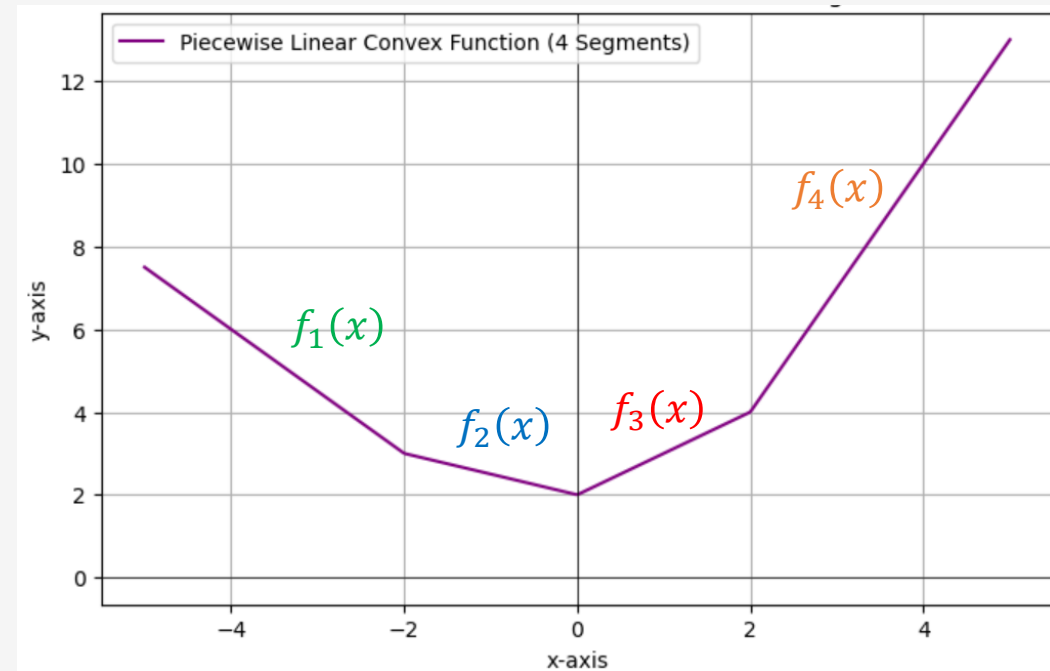
$$f(x) = \max_{i=1\dots m} \{f_i(x)\} = \max_{i=1\dots m} \{a_i^T x + b_i\}$$

$$f_1(x) = -(3/2)x,$$

$$f_2(x) = -(1/2)x + 2,$$

$$f_3(x) = x + 2,$$

$$f_4(x) = 3x - 2,$$



Minimization of convex piecewise linear functions

$$\min \max_{i=1\dots m} \{a_i^T x + b_i\}$$

Define a **new variable** $t \in \mathbb{R}$ such that

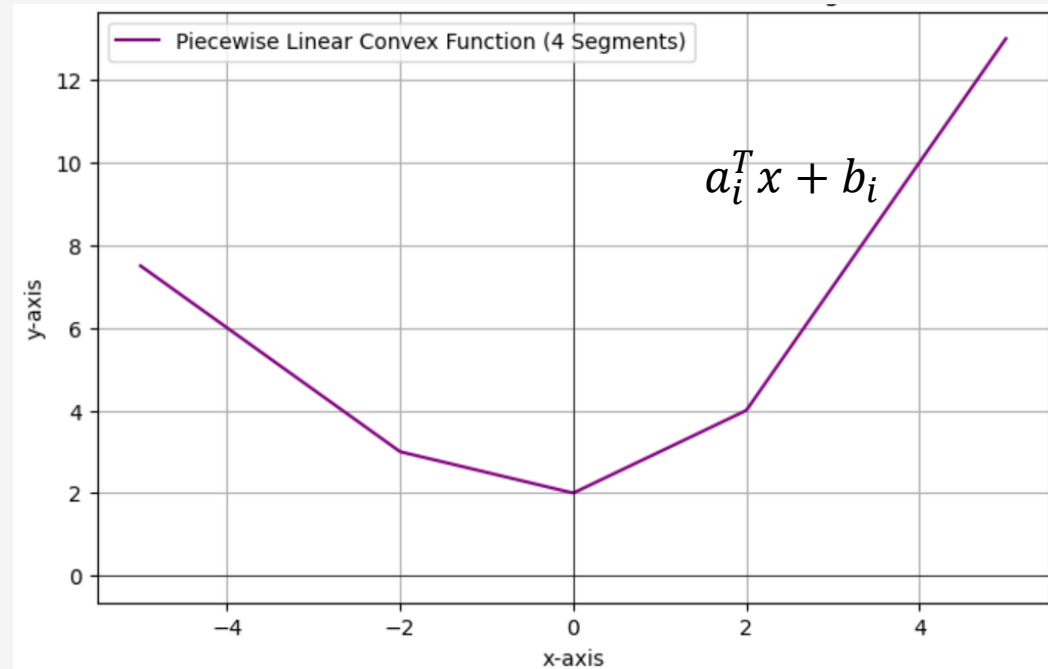
$$t = \max_{i=1\dots n} \{a_i^T x + b_i\}$$

Equivalent LP problem

$$\min t$$

$$\text{s.t. } a_i^T x + b_i \leq t, \quad i = 1, \dots, m$$

Objective function is **not differentiable everywhere**.



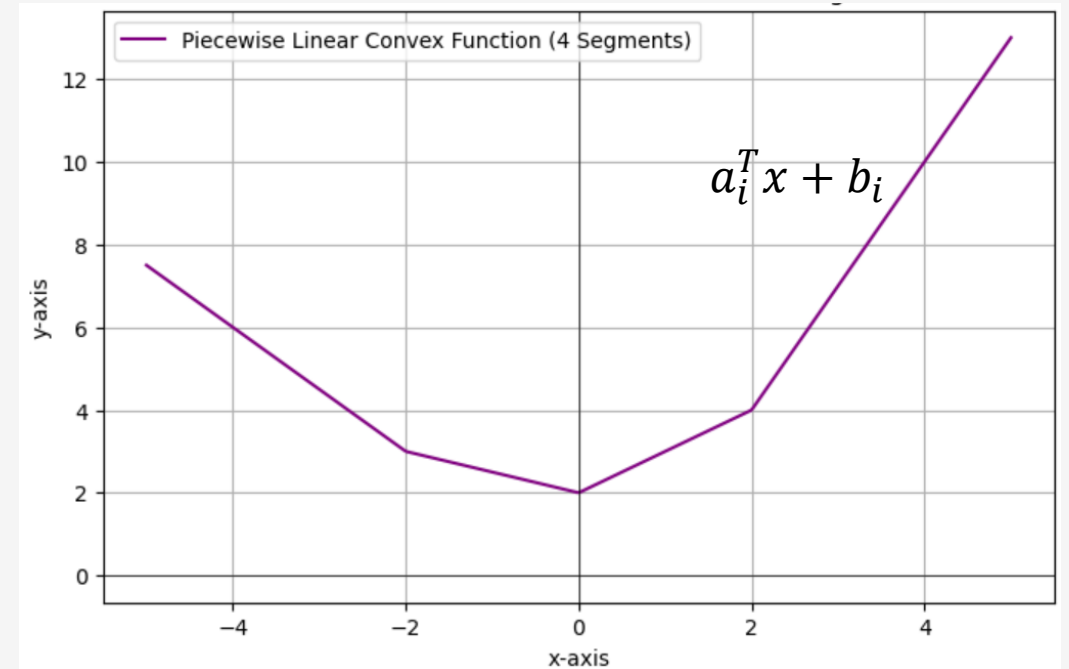
Equivalent LP problem

$$\begin{aligned} \min \quad & t \\ \text{s.t.} \quad & a_i^T x + b_i \leq t, \quad i = 1, \dots, m \end{aligned}$$

Matrix notation

$$\begin{aligned} \min \quad & \tilde{c}^T \tilde{x} \\ \text{s.t.} \quad & \tilde{A} \tilde{x} \leq \tilde{b}, \end{aligned}$$

$$\tilde{x} = \begin{pmatrix} x \\ t \end{pmatrix}, \quad \tilde{c} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \tilde{A} = \begin{bmatrix} a_1^T & -1 \\ \vdots & \vdots \\ a_m^T & -1 \end{bmatrix}, \quad \tilde{b} = \begin{pmatrix} -b_1 \\ \vdots \\ -b_m \end{pmatrix}$$



Vector norm problems as LPs

L_∞ -norm regression

(Minimize the maximum absolute residual)

Solve the L_∞ **optimization problem**: $\min \|Ax - b\|_\infty$

Recall the definition of the **absolute value**: $\|y\|_\infty = \max_{i=1..m} \{|y_i|\} = \max_{i=1..m} \{\max\{y_i, -y_i\}\}$

New variable: $t = \max_{i=1..m} \{|y_i|\}$, and rewrite it as m constraints: $t \geq y_i, \quad t \geq -y_i, \quad i = 1 \dots m$

Equivalent LP for the L_∞ -norm regression

$$\min \|Ax - b\|_\infty = \min \max_{i=1..m} \{|(Ax - b)_i|\} = \min \max_{i=1..m} \{\max\{(Ax - b)_i, -(Ax - b)_i\}\}$$

$$\begin{aligned} \min \quad & t \\ \text{s.t.} \quad & (Ax - b)_i \leq t, \quad i = 1 \dots m \\ & -(Ax - b)_i \leq t, \quad i = 1 \dots m \end{aligned}$$



$$\begin{aligned} \min \quad & t \\ \text{s.t.} \quad & Ax - b \leq t\mathbf{1} \\ & -(Ax - b) \leq t\mathbf{1} \end{aligned}$$

L_∞ -norm regression

Solve the L_∞ **optimization problem**:

$$\min \|Ax - b\|_\infty$$

Recall the definition of the absolute value:

$$\|y\|_\infty = \max_{i=1..m} \{|y_i|\} = \max_{i=1..m} \{\max\{y_i, -y_i\}\}$$

Equivalent LP for the L_∞ -norm regression – in matrix/vector form:

$$\begin{array}{ll}\min & t \\ & Ax - b \leq t\mathbf{1} \\ & -(Ax - b) \leq t\mathbf{1}\end{array}$$



$$\begin{array}{ll}\min & t \\ \text{s.t.} & Ax - b \leq t\mathbf{1} \\ & -Ax + b \leq t\mathbf{1}\end{array}$$



$$\begin{array}{ll}\min & \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix}^T \begin{pmatrix} x \\ t \end{pmatrix} \\ \text{s.t.} & \begin{bmatrix} A & -\mathbf{1} \\ -A & -\mathbf{1} \end{bmatrix} \begin{pmatrix} x \\ t \end{pmatrix} \leq \begin{pmatrix} b \\ -b \end{pmatrix}\end{array}$$

Sum of piecewise linear functions

Consider two piecewise linear functions: $f(x) = \max_{i=1..m} (a_i^T x + b_i)$ and $g(x) = \max_{i=1..p} (c_i^T x + d_i)$

We need to solve the minimization problem:

$$\min(f(x) + g(x)) = \min\left(\max_{i=1..m} (a_i^T x + b_i) + \max_{i=1..p} (c_i^T x + d_i)\right)$$

Equivalent LP

$$\begin{aligned} \min \quad & t_1 + t_2 \\ \text{s.t.} \quad & a_i^T x + b_i \leq t_1, i = 1 \dots m \\ & c_i^T x + d_i \leq t_2, i = 1 \dots p \end{aligned}$$

Exercise:
Write it in matrix/vector form

L_1 -norm regression (*Least Absolute Deviation - LAD*)

$$\min \|Ax - b\|_1 \rightarrow \min \sum_{i=1}^m |(Ax - b)_i|$$

In general, we can rewrite the L_1 -norm of any vector $y \in R^m$ as follows:

$$\|y\|_1 = \sum_{i=1}^m |y_i| = \sum_{i=1}^m \max\{y_i, -y_i\}$$

Equivalent LP

Define: $u_i = \max\{(Ax - b)_i, -(Ax - b)_i\}, \quad i = 1, \dots, m$

$\begin{aligned} \min \quad & u_1 + \dots + u_m \\ \text{s.t.} \quad & (Ax - b)_i \leq u_i, \quad i = 1 \dots m \\ & -(Ax - b)_i \leq u_i, \quad i = 1 \dots m \end{aligned}$		$\begin{aligned} \min \quad & \mathbf{1}^T u \\ \text{s.t.} \quad & Ax - b \leq u \\ & -(Ax - b) \leq u \end{aligned}$ where, $u = (u_1, \dots, u_m)^T$		$\begin{aligned} \min \quad & \begin{pmatrix} 0 \\ \mathbf{1} \end{pmatrix}^T \begin{pmatrix} x \\ u \end{pmatrix} \\ \text{s.t.} \quad & \begin{bmatrix} A & -I \\ -A & -I \end{bmatrix} \begin{pmatrix} x \\ u \end{pmatrix} \leq \begin{pmatrix} b \\ -b \end{pmatrix} \end{aligned}$ where, $I \in R^{m \times m}$ identity matrix
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Summary of L_∞ -norm and L_1 -norm regression

L_∞ -norm

$$\min \|Ax - b\|_\infty$$

Equivalent LP

$$\begin{array}{ll}\min & t \\ \text{s.t.} & Ax - b \leq t\mathbf{1} \\ & -Ax - b \leq t\mathbf{1}\end{array}$$

Remark: the absolute value of every element $(Ax - b)_i$ is bounded by the **same scalar** t

L_1 -norm

$$\min \|Ax - b\|_1$$

Equivalent LP

$$\begin{array}{ll}\min & \mathbf{1}^T u \\ \text{s.t.} & Ax - b \leq u \\ & -(Ax - b) \leq u\end{array}$$

Remark: the absolute value of every element $(Ax - b)_i$ is bounded by the elements of the **vector** $u = (u_1, \dots, u_m)^T$

Exercise (take-home...)

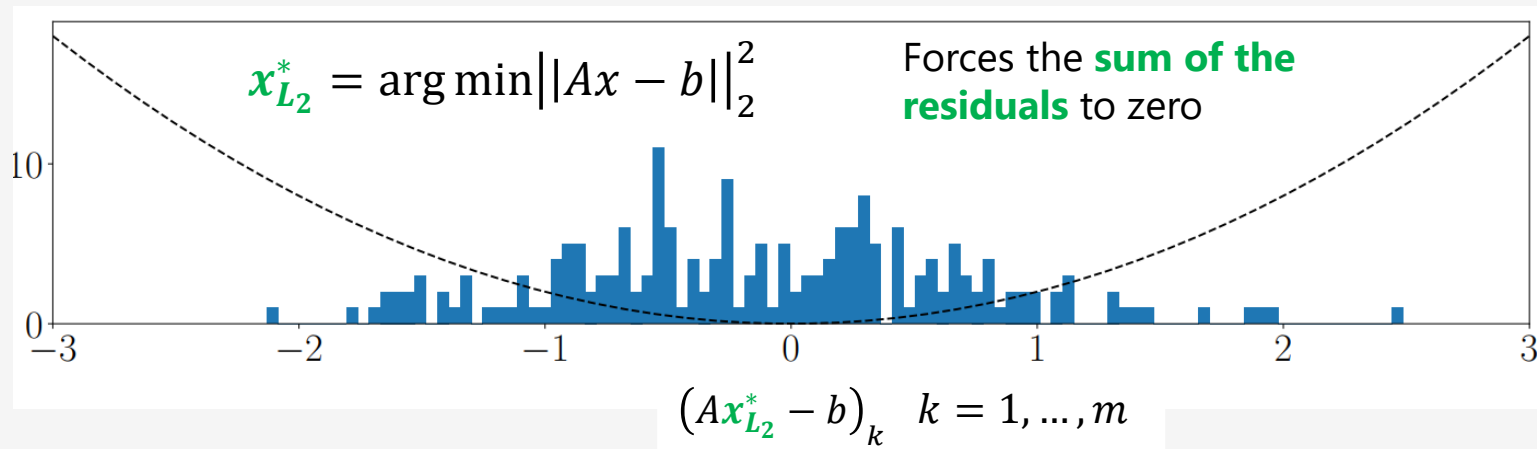
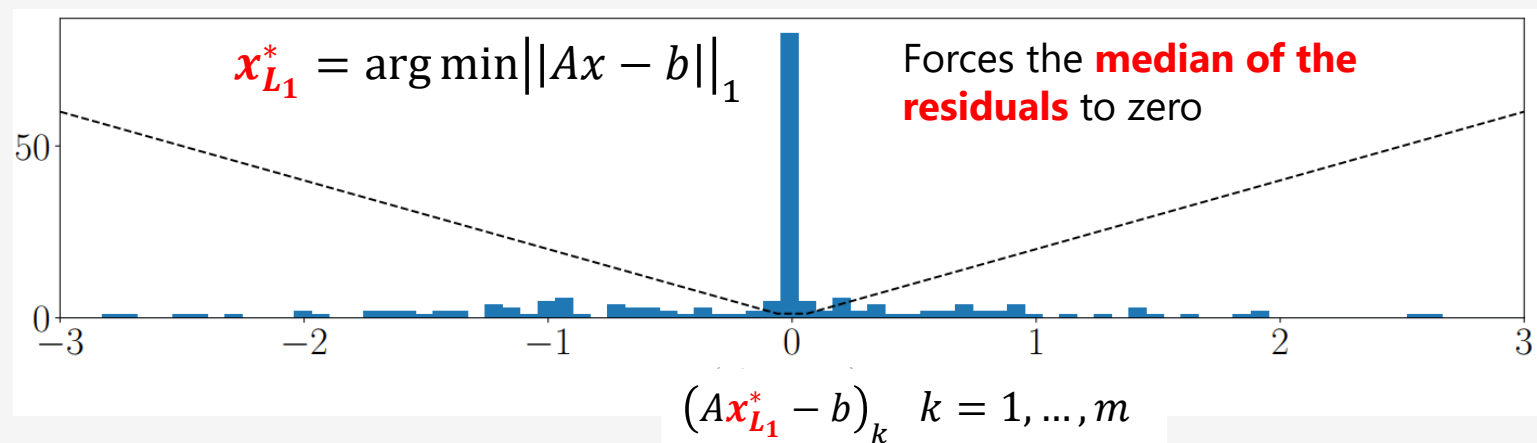
Convert the optimization problem to an equivalent LP

$$\min \|Ax - b\|_{\infty}$$

$$\text{s.t. } \|x\|_1 \leq k$$

Comparison of L_1 -norm regression with least squares (L_2 -norm)

Distribution of the residuals $(Ax - b)_k$ for $k = 1, \dots, m$ with randomly generated matrix $A \in R^{200 \times 80}$



Remark: the distribution of the L_1 -norm is wider with a high peak (concentration) at zero.

CVXPY makes easy to model & solve L_1 and L_∞ regression

L_∞ -norm

$$\min \|Ax - b\|_\infty$$

```
import numpy as np
import cvxpy as cp

m = 200; n = 80

A = np.random.randn(200, 80)
b = np.random.randn(200)
x = cp.Variable(80)

objective = cp.norm(A @ x - b, np.inf)
problem = cp.Problem(cp.Minimize(objective))
problem.solve()
```

L_1 -norm

$$\min \|Ax - b\|_1$$

```
import numpy as np
import cvxpy as cp

m = 200; n = 80

A = np.random.randn(200, 80)
b = np.random.randn(200)
x = cp.Variable(80)

objective = cp.norm(A @ x - b, 1)
problem = cp.Problem(cp.Minimize(objective))
problem.solve()
```

Sparse signal recovery

Sparse signal recovery via L_1 -norm minimization

$\hat{x} \in R^n$ is an **unknown signal**

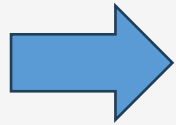
Prior knowledge: $\hat{x}$ is a sparse signal (i.e., most of its elements are zero)

We make linear measurements $y = A\hat{x}$ with known $A \in R^{m \times n}$ and $m < n$

Goal: Estimate signal $\hat{x}$ using L_1 -norm minimization (optimal solution has most elements zeros)

$$\begin{array}{ll} \min & \|x\|_1 \\ \text{s.t.} & Ax = y \end{array}$$

Equivalent LP



$$\begin{array}{ll} \min & \mathbf{1}^T u \\ \text{s.t.} & Ax = y \\ & -u \leq x \leq u \end{array}$$

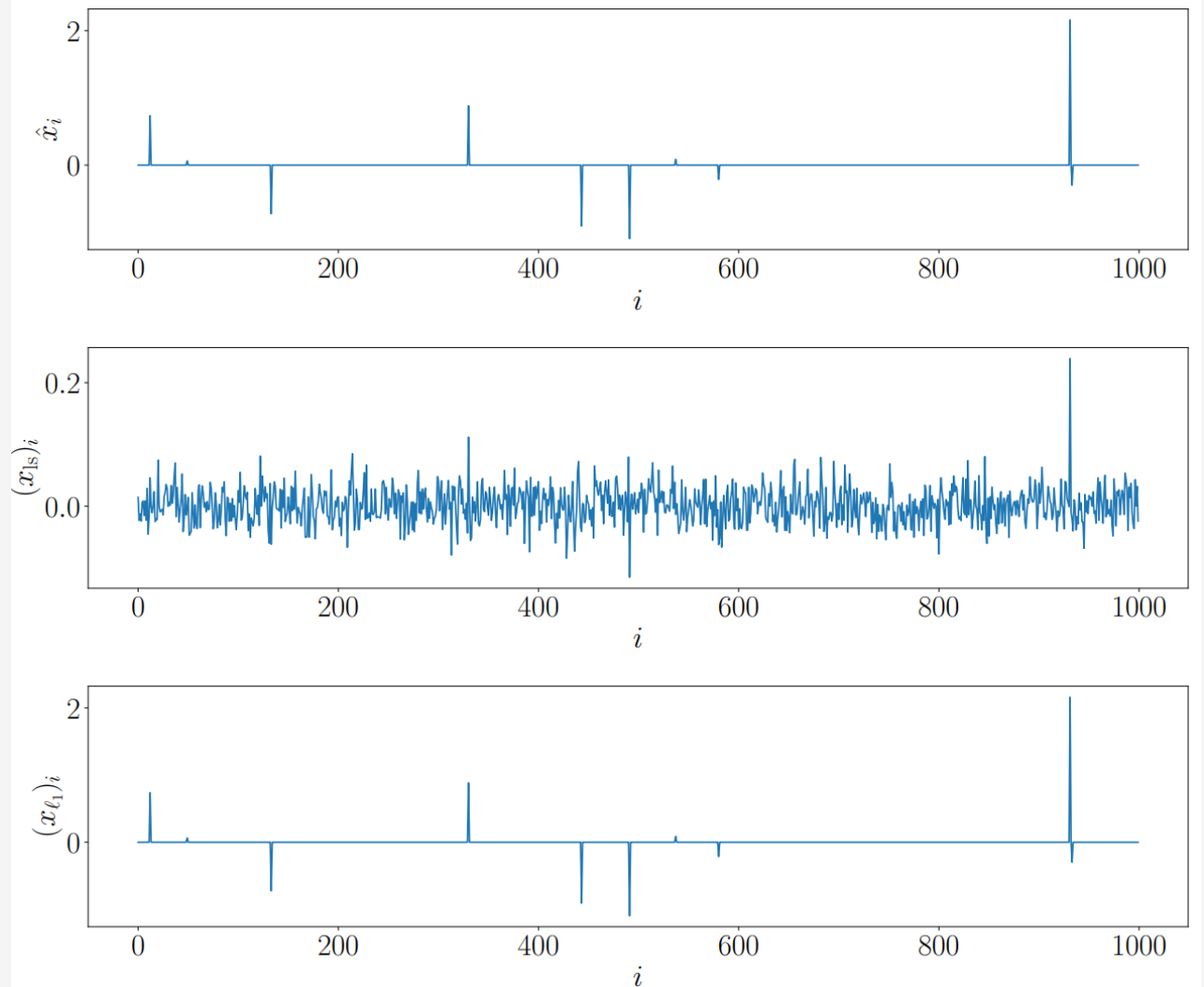
where $u = (u_1, \dots, u_n)^T$

Sparse signal recovery via ℓ_1 -norm minimization

Example: Exact signal $\hat{x} \in \mathbb{R}^{1000}$
10 nonzero components
Random $A \in \mathbb{R}^{100 \times 1000}$

The **least squares (L_2) estimate**
cannot recover the sparse signal
(No prior knowledge is included)

The **L_1 -norm estimate** is able to
recover the **exact** sparse signal
(Our prior knowledge is included)



Support Vector Machines

SVMs for linear classification (linear separation)

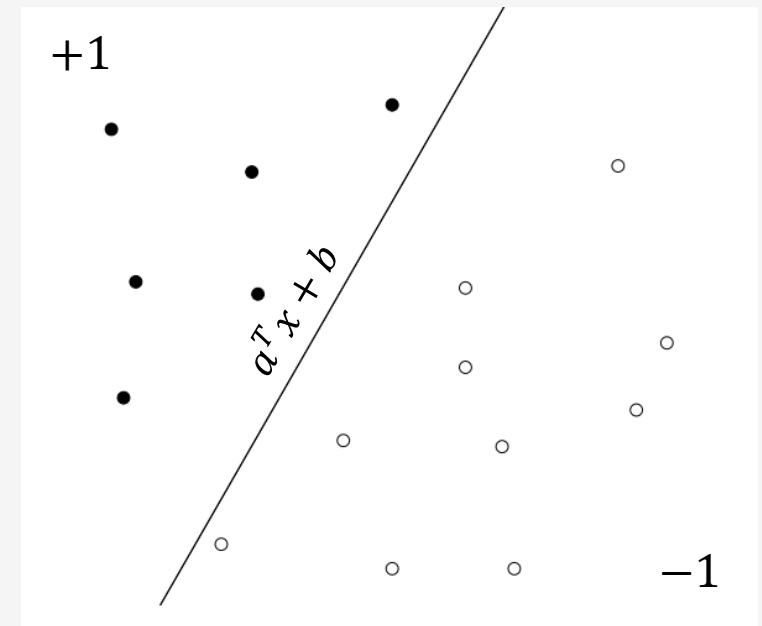
Data: Given a set of points $V = \{v_1, \dots, v_N\}$ and their corresponding labels $S = \{s_1, \dots, s_N\}$ with $s_i \in \{-1, 1\}$

Goal: Find a hyperplane $a^T x + b$ that strictly separates the two classes

$$\begin{aligned} a^T v_i + b &> 0, \text{ if } s_i = +1 \\ a^T v_i + b &< 0, \text{ if } s_i = -1 \end{aligned} \quad i = 1, \dots, N$$

Since the affine function $a^T x + b$ is **homogeneous** in (a, b) , we can write the two inequalities as follows:

$$s_i(a^T v_i + b) \geq 1 \quad i = 1, \dots, N$$



SVMs for linear classification (*sparse case*)

Feasibility problem

$$\begin{array}{ll}\text{find} & a, b \\ \text{s.t.} & s_i(a^T v_i + b) \geq 1, i = 1 \dots N\end{array}$$

Equivalent LP

$$\begin{array}{ll}\text{min} & 0 \\ \text{s.t.} & s_i(a^T v_i + b) \geq 1, i = 1 \dots N\end{array}$$

Case #1: if $p^* = 0$ then the problem is **feasible** $\Rightarrow \exists a, b \Rightarrow$ points v_i are **separable**

Case #2: if $p^* = \infty$ then the problems is **infeasible** $\Rightarrow \nexists a, b \Rightarrow$ points v_i are **NOT separable**

...needs more work $\rightarrow$

SVMs for linear classification (*non-separable points*)

Some of the constraints may be violated (**misclassified points**)

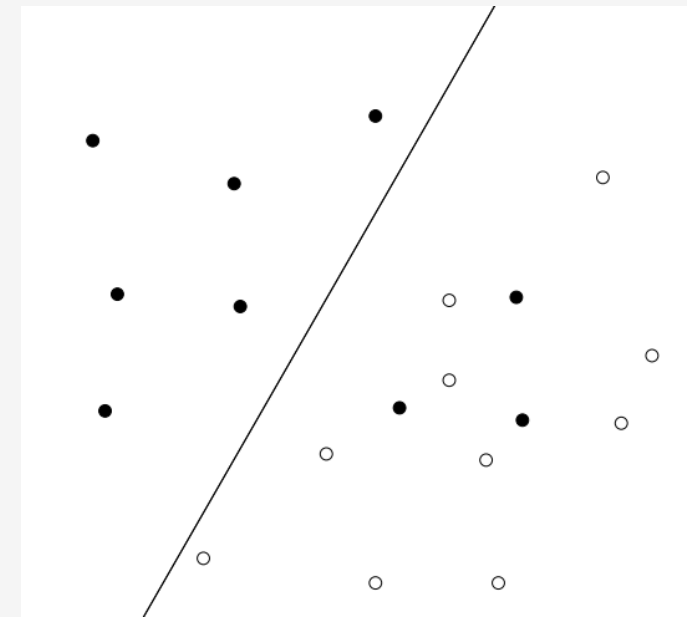
$$s_i(a^T v_i + b) \geq 1 \quad \rightarrow \quad \max\{0, 1 - s_i(a^T v_i + b)\}$$

Goal: minimize the sum of the violations

$$\min \sum_{i=1}^N \max\{0, 1 - s_i(a^T v_i + b)\}$$

Piecewise-linear minimization problem with variables a, b

We can convert it to an LP



References

D. Bertsimas & J. Tsitsiklis: Introduction to Linear Optimization
Chapter 1.3: piecewise linear optimization

R. Vanderbei: Linear Programming – Foundations & Extensions
Chapter 12.4, 12.7: 1-norm regression and SVMs

Next lecture

- Linear optimization geometry
- Optimality conditions
